

Chemical Coordination and Integration

The neural system provides a point-to-point rapid coordination among organs. The neural coordination is fast but short-lived.

As the nerve fibres do not innervate all cells of the body and the cellular functions need to be continuously regulated, a special kind of coordination and integration has to be provided. This function is carried out by hormones.

The neural system and the endocrine system jointly coordinate and regulate the physiological functions in the body.

[MEMORY AID - not in NCERT] Two coordinating systems, two speeds: neural = fast but short-lived and point-to-point; endocrine = slower but sustained and broadcast in the blood. Every fact in this chapter hangs on that contrast.

19.1 ENDOCRINE GLANDS AND HORMONES

- Endocrine glands lack ducts and are hence called ductless glands. Their secretions are called hormones.

Gland type	Duct	Where the secretion goes
Endocrine (ductless)	Absent	Released as a hormone , carried in the blood
Exocrine	Present	Poured through the duct on to a surface or into a cavity

[NOTE] Exocrine gland - the chapter uses the word without defining it (the pancreas acts as both exocrine and endocrine gland, and secretin acts on the exocrine pancreas), so the contrast is stated here from the chapter's own words only: an endocrine gland is ductless, therefore an exocrine gland is the duct-bearing kind. Nothing beyond that is claimed.

The classical definition of hormone as a chemical produced by endocrine glands and released into the blood and transported to a distantly located target organ has a current scientific definition as follows.

- Hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts.

The new definition covers a number of new molecules in addition to the hormones secreted by the organised endocrine glands.

- Invertebrates possess very simple endocrine systems with few hormones, whereas a large number of chemicals act as hormones and provide coordination in the vertebrates.
- The human endocrine system is described here.
- These hormones regulate metabolism, growth and development of our organs, the endocrine glands or certain cells.

19.2 HUMAN ENDOCRINE SYSTEM

- The endocrine glands and hormone producing diffused tissues/cells located in different parts of our body constitute the endocrine system.

Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females) are the organised endocrine bodies in our body (Figure 19.1). Counting the hypothalamus, the endocrine system is composed of hypothalamus, pituitary and pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis and ovary).

In addition to these, some other organs, e.g., gastrointestinal tract, liver, kidney, heart also produce hormones. A brief account of the structure and functions of all major endocrine glands and hypothalamus of the human body is given in the following sections.

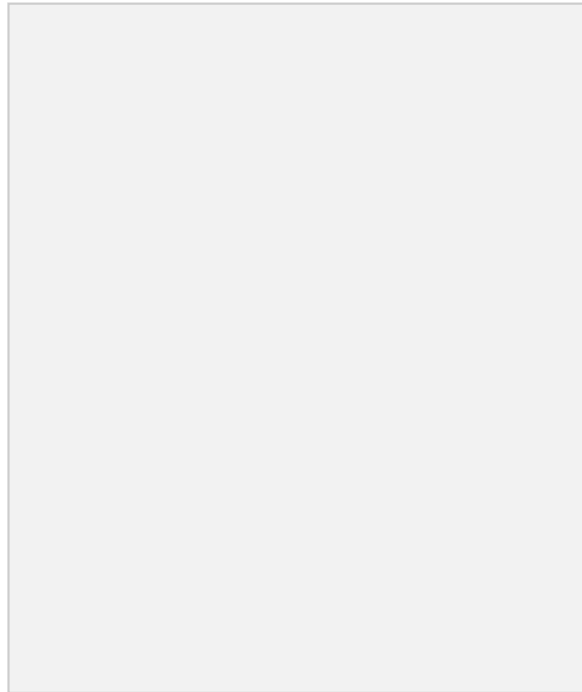


Figure 19.1 Location of endocrine glands

[MEMORY AID - not in NCERT] Ten glands, head to gonad, in the order Section 19.2 uses them: Hypothalamus, Pituitary, Pineal, Thyroid, Parathyroid, Thymus, Adrenal, Pancreas, Testis, Ovary. Learn the order once and every subsection number becomes predictable.

19.2.1

The Hypothalamus

The hypothalamus is the basal part of diencephalon, forebrain (Figure 19.1) and it regulates a wide spectrum of body functions.

- It contains several groups of neurosecretory cells called nuclei which produce hormones. These hormones regulate the synthesis and secretion of pituitary hormones.

Hypothalamic hormone type	What it does	Example
Releasing hormones	Stimulate secretion of pituitary hormones	Gonadotrophin releasing hormone (GnRH) stimulates the pituitary synthesis and release of gonadotrophins
Inhibiting hormones	Inhibit secretions of pituitary hormones	Somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary

- These hormones originate in the hypothalamic neurons.
- They pass through axons and are released from their nerve endings.
- They reach the pituitary gland through a portal circulatory system.
- There they regulate the functions of the anterior pituitary.

[NOTE] The **posterior pituitary** is under the **direct neural regulation of the hypothalamus** (Figure 19.2) - so the two lobes are reached by two different routes: the **anterior** by **blood** (portal system), the **posterior** by **nerve**.

19.2.2

The Pituitary Gland

The pituitary gland is located in a **bony cavity called sella tursica** (the source's own spelling) and is **attached to hypothalamus by a stalk** (Figure 19.2). It is divided anatomically into an **adenohypophysis** and a **neurohypophysis**.

Anatomical division	Part	Hormones
Adenohypophysis - consists of two portions	Pars distalis, commonly called anterior pituitary	Growth hormone (GH), prolactin (PRL), thyroid stimulating hormone (TSH), adrenocorticotrophic hormone (ACTH), luteinizing hormone (LH) and follicle stimulating hormone (FSH) - six trophic hormones
	Pars intermedia	Secretes only one hormone called melanocyte stimulating hormone (MSH) ; in humans the pars intermedia is almost merged with pars distalis
Neurohypophysis (pars nervosa)	Also known as posterior pituitary	Stores and releases two hormones, oxytocin and vasopressin , which are actually synthesised by the hypothalamus and are transported axonally to neurohypophysis

[NOTE] The pituitary gland is divided into **three major parts**, which are called as **pars distalis, pars intermedia** and **pars nervosa**. Two of the three (distalis, intermedia) make the adenohypophysis; the third is the neurohypophysis.

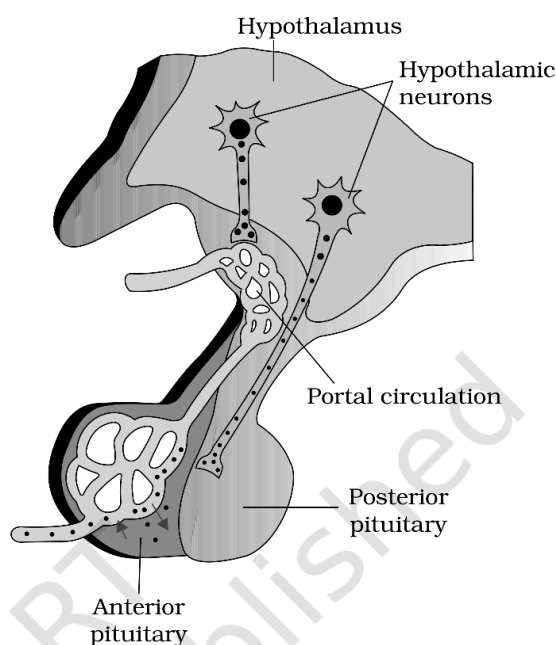


Figure 19.2 Diagrammatic representation of pituitary and its relationship with hypothalamus

Growth hormone (GH) - the two directions of error

Condition	Secretion	Effect
Gigantism	Over-secretion of GH	Stimulates abnormal growth of the body
Pituitary dwarfism	Low secretion of GH	Stunted growth
Acromegaly	Excess secretion of growth hormone in adults, especially in middle age	Severe disfigurement (especially of the face) , which may lead to serious complications and premature death if unchecked

[NOTE] Acromegaly is hard to diagnose in the early stages and often goes undetected for many years, until changes in external features become noticeable.

What each anterior-pituitary hormone does

Hormone	Action
Prolactin (PRL)	Regulates the growth of the mammary glands and formation of milk in them
TSH - the exercises call it Thyrotrophin (TSH)	Stimulates the synthesis and secretion of thyroid hormones from the thyroid gland
ACTH - the exercises call it Corticotrophin (ACTH)	Stimulates the synthesis and secretion of steroid hormones called glucocorticoids from the adrenal cortex
LH and FSH	Stimulate gonadal activity and hence are called gonadotrophins
LH in males	Stimulates the synthesis and secretion of hormones called androgens from testis
FSH in males	FSH and androgens regulate spermatogenesis
LH in females	Induces ovulation of fully mature follicles (graafian follicles) and maintains the corpus luteum, formed from the remnants of the graafian follicles after ovulation
FSH in females	Stimulates growth and development of the ovarian follicles
MSH - the exercises call it Melanotrophin (MSH)	Acts on the melanocytes (melanin containing cells) and regulates pigmentation of the skin

The two posterior-pituitary hormones

Hormone	Action
Oxytocin	Acts on the smooth muscles of our body and stimulates their contraction . In females it stimulates a vigorous contraction of uterus at the time of child birth , and milk ejection from the mammary gland
Vasopressin	Acts mainly at the kidney and stimulates resorption of water and electrolytes by the distal tubules , thereby reducing loss of water through urine (diuresis) ; hence it is also called anti-diuretic hormone (ADH)

[NOTE] An impairment affecting synthesis or release of ADH results in a **diminished ability of the kidney to conserve water**, leading to **water loss and dehydration**. This condition is known as **Diabetes Insipidus** - a water disorder, not a sugar disorder.

Taken together, the **pituitary hormones regulate the growth and development of somatic tissues and the activities of peripheral endocrine glands**.

[MEMORY AID - not in NCERT] Six from pars distalis: GH, PRL, TSH, ACTH, LH, FSH - "Good People Take All Life's Favours". One from pars intermedia: MSH. Two stored in pars nervosa: oxytocin, vasopressin. 6 + 1 + 2.

19.2.3

The Pineal Gland

The pineal gland is located on the **dorsal side of forebrain**. Pineal secretes a hormone called **melatonin**.

- Melatonin plays a **very important role in the regulation of a 24-hour (diurnal) rhythm** of our body.
 - For example, it helps in **maintaining the normal rhythms of sleep-wake cycle, body temperature**.

- In addition, melatonin also influences **metabolism, pigmentation, the menstrual cycle** as well as our **defense capability**.

19.2.4

Thyroid Gland

The thyroid gland is composed of **two lobes** which are located on **either side of the trachea** (Figure 19.3 a). Both the lobes are **interconnected with a thin flap of connective tissue called isthmus**.

- The thyroid gland is composed of **follicles and stromal tissues**. Each **thyroid follicle** is composed of **follicular cells, enclosing a cavity**.

These **follicular cells synthesise two hormones, tetraiodothyronine or thyroxine (T_4) and triiodothyronine (T_3)**. Iodine is **essential for the normal rate of hormone synthesis in the thyroid**.

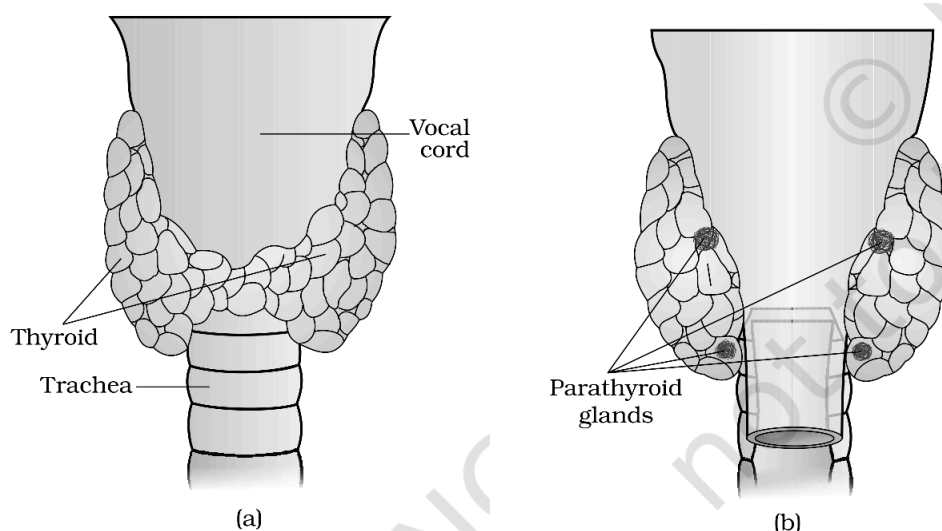


Figure 19.3 Diagrammatic view of the position of Thyroid and Parathyroid (a) Ventral side (b) Dorsal side

[NOTE] Reading Figure 19.3: On the ventral side (a), the **vocal cord, thyroid, and trachea** are shown. On the dorsal side (b), the four **parathyroid glands** are embedded on the posterior surface of the thyroid lobes.

Too little iodine, too much hormone

Disorder	Cause	Consequences
Goitre with hypothyroidism	Deficiency of iodine in our diet	Hypothyroidism and enlargement of the thyroid gland , commonly called goitre
Cretinism	Hypothyroidism during pregnancy	Defective development and maturation of the growing baby leading to stunted growth (cretinism), mental retardation, low intelligence quotient, abnormal skin, deaf-mutism, etc.
Irregular cycle	Hypothyroidism in adult women	The menstrual cycle may become irregular
Hyperthyroidism	Cancer of the thyroid gland or development of nodules of the thyroid glands	Rate of synthesis and secretion of the thyroid hormones is increased to abnormal high levels , which adversely affects the body physiology
Exophthalmic goitre (the source's own spelling), also called Graves' disease	A form of hyperthyroidism	Enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate and weight loss

What the thyroid hormones do

- Thyroid hormones play an important role in the **regulation of the basal metabolic rate**.
- These hormones also **support the process of red blood cell formation**.
- Thyroid hormones **control the metabolism of carbohydrates, proteins and fats**.
- **Maintenance of water and electrolyte balance** is also influenced by thyroid hormones.
- Collected in one line: the thyroid gland hormones play an important role in the regulation of the **basal metabolic rate, development and maturation of the central neural system, erythropoiesis, metabolism of carbohydrates, proteins and fats, menstrual cycle**.
- Thyroid gland **also secretes a protein hormone called thyrocalcitonin (TCT)** which **regulates the blood calcium levels** - thyrocalcitonin regulates calcium levels in our blood **by decreasing it**.

19.2.5

Parathyroid Gland

In humans, **four parathyroid glands** are present on the **back side of the thyroid gland**, **one pair each in the two lobes** of the thyroid gland (Figure 19.3 b).

- The parathyroid glands secrete a **peptide hormone called parathyroid hormone (PTH)**. The secretion of PTH is **regulated by the circulating levels of calcium ions**.

- | | |
|---|---|
| 1 | Parathyroid hormone (PTH) increases the Ca^{2+} levels in the blood. |
| 2 | PTH acts on bones and stimulates the process of bone resorption (dissolution/demineralisation) . |
| 3 | PTH also stimulates reabsorption of Ca^{2+} by the renal tubules . |
| 4 | PTH increases Ca^{2+} absorption from the digested food. |

- It is, thus, clear that **PTH is a hypercalcemic hormone**, i.e., it **increases the blood Ca^{2+} levels**. Along with TCT, it plays a **significant role in calcium balance** in the body.

[MEMORY AID - not in NCERT] Calcium see-saw: PTH pushes calcium UP (hypercalcemic: bone, kidney, gut all deliver calcium to the blood); TCT pulls it DOWN. Same ion, opposite glands - thyroid down, parathyroid up.

19.2.6

Thymus

The thymus gland is a **lobular structure located between lungs behind sternum on the ventral side of aorta**. The **thymus plays a major role in the development of the immune system**.

- This gland secretes the **peptide hormones called thymosins**.
- Thymosins play a **major role in the differentiation of T-lymphocytes**, which provide **cell-mediated immunity**.
- In addition, thymosins also **promote production of antibodies** to provide **humoral immunity**.
- **Thymus is degenerated in old individuals** resulting in a **decreased production of thymosins**. As a result, the **immune responses of old persons become weak**.

Our body has one pair of adrenal glands, one above of each kidney (Figure 19.4 a). The gland is composed of two types of tissues: the centrally located tissue is called the adrenal medulla, and outside this lies the adrenal cortex (Figure 19.4 b).

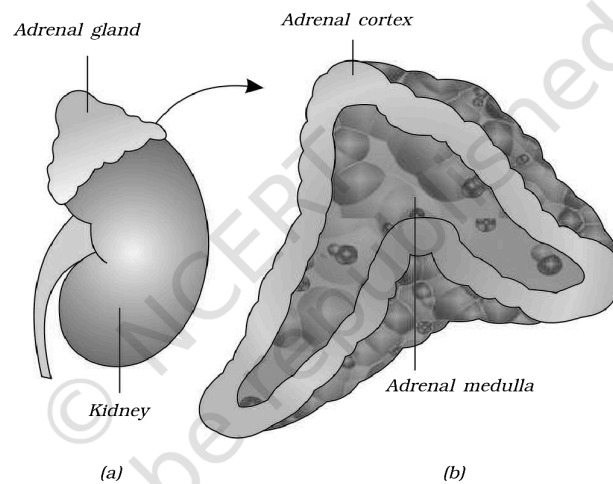


Figure 19.4 Diagrammatic representation of : (a) Adrenal gland above kidney (b) Section showing two parts of adrenal gland

[NOTE] Underproduction of hormones by the adrenal cortex alters carbohydrate metabolism, causing acute weakness and fatigue, leading to a disease called Addison's disease.

Medulla

Adrenal medulla - the catecholamines

- The adrenal medulla secretes two hormones called **adrenaline or epinephrine** and **noradrenaline or norepinephrine**. These are commonly called as **catecholamines**.

Adrenaline and noradrenaline are **rapidly secreted in response to stress of any kind and during emergency situations** and are called **emergency hormones** or **hormones of Fight or Flight**.

Catecholamine effect	What happens
Alertness and skin	These hormones increase alertness , pupillary dilation (the source's own spelling), piloerection (raising of hairs), sweating etc.
Heart and lungs	Both the hormones increase the heart beat , the strength of heart contraction and the rate of respiration
Carbohydrate	Catecholamines stimulate the breakdown of glycogen resulting in an increased concentration of glucose in blood
Fat and protein	In addition, they also stimulate the breakdown of lipids and proteins

[NOTE] The one-line list to reproduce in an exam: these hormones increase **alertness**, **pupillary dilation**, **piloerection**, **sweating**, **heart beat**, **strength of heart contraction**, **rate of respiration**, **glycogenolysis**, **lipolysis**, **proteolysis**. The last three are the **metabolic names** for the breakdown of glycogen, lipids and proteins above.

The adrenal cortex can be divided into **three layers**, called **zona reticularis (inner layer)**, **zona fasciculata (middle layer)** and **zona glomerulosa (outer layer)**. The adrenal cortex secretes many hormones, commonly called as **corticoids**.

Corticoid class	Definition	Main one in our body
Glucocorticoids	The corticoids involved in carbohydrate metabolism	Cortisol is the main glucocorticoid
Mineralocorticoids	Corticoids which regulate the balance of water and electrolytes in our body	Aldosterone is the main mineralocorticoid in our body

- Glucocorticoids stimulate gluconeogenesis, lipolysis and proteolysis; and inhibit cellular uptake and utilisation of amino acids.
- Cortisol is also involved in maintaining the cardio-vascular system as well as the kidney functions.
- Glucocorticoids, particularly cortisol, produce anti-inflammatory reactions and suppress the immune response.
- Cortisol stimulates the RBC production.
- Collected in one line: glucocorticoids stimulate gluconeogenesis, lipolysis, proteolysis, erythropoiesis, cardio-vascular system, blood pressure, and glomerular filtration rate, and inhibit inflammatory reactions by suppressing the immune response.

1	Aldosterone acts mainly at the renal tubules.
2	It stimulates the reabsorption of Na^+ and water.
3	It stimulates the excretion of K^+ and phosphate ions.
4	Thus aldosterone helps in the maintenance of electrolytes, body fluid volume, osmotic pressure and blood pressure.

[NOTE] Small amounts of androgenic steroids are also secreted by the adrenal cortex, which play a role in the growth of axial hair, pubic hair and facial hair during puberty.

[MEMORY AID - not in NCERT] Cortex from outside in - "salt, sugar, sex": zona glomerulosa (outer) is the salt layer, zona fasciculata (middle) the sugar layer, zona reticularis (inner) the sex-steroid layer. NCERT lists them inner-first, so read the list backwards.

Pancreas is a **composite gland** (Figure 19.1) which acts as **both exocrine and endocrine gland**.

- The **endocrine pancreas** consists of 'Islets of Langerhans'. There are about **1 to 2 million Islets of Langerhans** in a normal human pancreas, representing only **1 to 2 per cent** of the pancreatic tissue.

Islet cell	Hormone	Chemical nature
alpha-cells	Glucagon	Peptide hormone; plays an important role in maintaining the normal blood glucose levels
beta-cells	Insulin	Peptide hormone; plays a major role in the regulation of glucose homeostasis

[NOTE] The source prints these two cell names with the Greek letters **alpha** and **beta** before the word **cells**; they are spelled out here so the page stays a plain-ASCII print file. The **two main types of cells in the Islet of Langerhans** are exactly these two.

Glucagon - the hyperglycemic hormone

1	Glucagon acts mainly on the liver cells (hepatocytes) .
2	It stimulates glycogenolysis , resulting in an increased blood sugar (hyperglycemia) .
3	In addition, this hormone stimulates the process of gluconeogenesis , which also contributes to hyperglycemia .
4	Glucagon reduces the cellular glucose uptake and utilisation .
5	Thus, glucagon is a hyperglycemic hormone .

Insulin - the hypoglycemic hormone

1	Insulin acts mainly on hepatocytes and adipocytes (cells of adipose tissue) .
2	It enhances cellular glucose uptake and utilisation .
3	As a result, there is a rapid movement of glucose from blood to hepatocytes and adipocytes , resulting in decreased blood glucose levels (hypoglycemia) .
4	Insulin also stimulates conversion of glucose to glycogen (glycogenesis) in the target cells.

- The **glucose homeostasis in blood** is thus **maintained jointly by the two - insulin and glucagons** (the source's own plural spelling of glucagon).

[NOTE] Prolonged hyperglycemia leads to a complex disorder called **diabetes mellitus**, which is associated with **loss of glucose through urine and formation of harmful compounds known as ketone bodies**. **Insulin deficiency and/or insulin resistance** result in this disease, and **diabetic patients are successfully treated with insulin therapy**.

[MEMORY AID - not in NCERT] Alphabet trick: Alpha comes first and gives glucAgon, which raises sugar; Beta gives insulin, which Brings sugar down. Hyper- and hypo-glycemia are therefore alpha and beta respectively.

19.2.9

Testis

A **pair of testis** is present in the **scrotal sac (outside abdomen)** of male individuals (Figure 19.1). **Testis performs dual functions as a primary sex organ as well as an endocrine gland**.

- Testis is composed of **seminiferous tubules and stromal or interstitial tissue**. The **Leydig cells or interstitial cells**, which are present in the **intertubular spaces**, produce a group of hormones called **androgens**, mainly **testosterone**.
- Androgens **regulate the development, maturation and functions of the male accessory sex organs** like **epididymis, vas deferens, seminal vesicles, prostate gland, urethra** etc.

- These hormones **stimulate muscular growth**, **growth of facial and axillary hair**, **aggressiveness**, **low pitch of voice** etc.
- Androgens play a **major stimulatory role** in the process of **spermatogenesis** (formation of spermatozoa).
- Androgens **act on the central neural system** and **influence the male sexual behaviour** (libido).
- These hormones produce **anabolic (synthetic) effects** on **protein and carbohydrate metabolism**.
- Collected in one line: the testis secretes androgens, which stimulate the **development, maturation and functions of the male accessory sex organs**, **appearance of the male secondary sex characters**, **spermatogenesis**, **male sexual behaviour**, **anabolic pathways** and **erythropoiesis**.

19.2.10

Ovary

Females have a **pair of ovaries** located in the **abdomen** (Figure 19.1). **Ovary is the primary female sex organ** which **produces one ovum during each menstrual cycle**. In addition, ovary also produces **two groups of steroid hormones** called **estrogen and progesterone**.

- Ovary is composed of **ovarian follicles** and **stromal tissues**. The **estrogen** is **synthesised and secreted** mainly by the **growing ovarian follicles**.

1	<i>(cycle - last step feeds back to step 1)</i>
2	A follicle grows and secretes estrogen .
3	Ovulation occurs.
4	After ovulation, the ruptured follicle is converted to a structure called corpus luteum .
5	The corpus luteum secretes mainly progesterone .

Hormone	Actions
Estrogens	Produce wide ranging actions such as stimulation of growth and activities of female secondary sex organs , development of growing ovarian follicles , appearance of female secondary sex characters (e.g., high pitch of voice , etc.) and mammary gland development . Estrogens also regulate female sexual behaviour
Progesterone - the exercises call it the Progestational hormone	Progesterone supports pregnancy. It also acts on the mammary glands and stimulates the formation of alveoli (sac-like structures which store milk) and milk secretion . Progesterone thus plays a major role in the maintenance of pregnancy as well as in mammary gland development and lactation

19.3

HORMONES OF HEART, KIDNEY AND GASTROINTESTINAL TRACT

Hormones are also secreted by some tissues which are not endocrine glands.

Non-gland source	Hormone	Action
Heart - the atrial wall	Atrial natriuretic factor (ANF) , a very important peptide hormone	Decreases blood pressure
Kidney - the juxtaglomerular cells	Erythropoietin , a peptide hormone	Stimulates erythropoiesis (formation of RBC)

1	Blood pressure is increased.
2	ANF is secreted.
3	This causes dilation of the blood vessels.
4	This reduces the blood pressure.

- Endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones, namely gastrin, secretin, cholecystokinin (CCK) and gastric inhibitory peptide (GIP).

GI hormone	Acts on	Action
Gastrin	The gastric glands	Stimulates the secretion of hydrochloric acid and pepsinogen
Secretin	The exocrine pancreas	Stimulates secretion of water and bicarbonate ions
CCK	Both pancreas and gall bladder	Stimulates the secretion of pancreatic enzymes and bile juice, respectively
GIP	The stomach	Inhibits gastric secretion and motility

[NOTE] Taken together, these hormones regulate the secretion of digestive juices and help in digestion - three stimulate a secretion and the fourth, GIP, shuts one down.

Several other non-endocrine tissues secrete hormones called growth factors. These factors are essential for the normal growth of tissues and their repairing/regeneration.

[MEMORY AID - not in NCERT] Four GI hormones, three organs, one inhibitor: Gastrin -> stomach acid, Secretin -> pancreatic bicarbonate, CCK -> pancreatic enzymes + bile, GIP -> the only one that inhibits. "GIP is the Grumpy Inhibitory Peptide."

19.4 MECHANISM OF HORMONE ACTION

Hormones produce their effects on target tissues by binding to specific proteins called hormone receptors located in the target tissues only.

Receptor class	Where it sits
Membrane-bound receptors	Hormone receptors present on the cell membrane of the target cells
Intracellular receptors	Receptors present inside the target cell, mostly nuclear receptors (present in the nucleus)

1	Binding of a hormone to its receptor leads to the formation of a hormone-receptor complex (Figure 19.5 a, b).
2	Each receptor is specific to one hormone only and hence receptors are specific.
3	Hormone-Receptor complex formation leads to certain biochemical changes in the target tissue.
4	Target tissue metabolism and hence physiological functions are regulated by hormones.

On the basis of their **chemical nature**, hormones can be divided into groups :

Group	Examples
(i) peptide, polypeptide, protein hormones	e.g., insulin, glucagon, pituitary hormones, hypothalamic hormones, etc.
(ii) steroids	e.g., cortisol, testosterone, estradiol and progesterone
(iii) iodothyronines	thyroid hormones
(iv) amino-acid derivatives	e.g., epinephrine

Receptor used	Does the hormone enter the cell?	What it regulates
Membrane-bound	Normally do not enter the target cell, but generate second messengers (e.g., cyclic AMP, IP ₃ , Ca ⁺⁺ etc)	The second messengers in turn regulate cellular metabolism (Figure 19.5a)
Intracellular (e.g., steroid hormones, iodothyronines, etc.)	Enter and bind inside the cell	Mostly regulate gene expression or chromosome function by the interaction of hormone-receptor complex with the genome

[NOTE] Cumulative biochemical actions result in physiological and developmental effects (Figure 19.5b) - which is why a single hormone-receptor binding event can end in tissue growth.

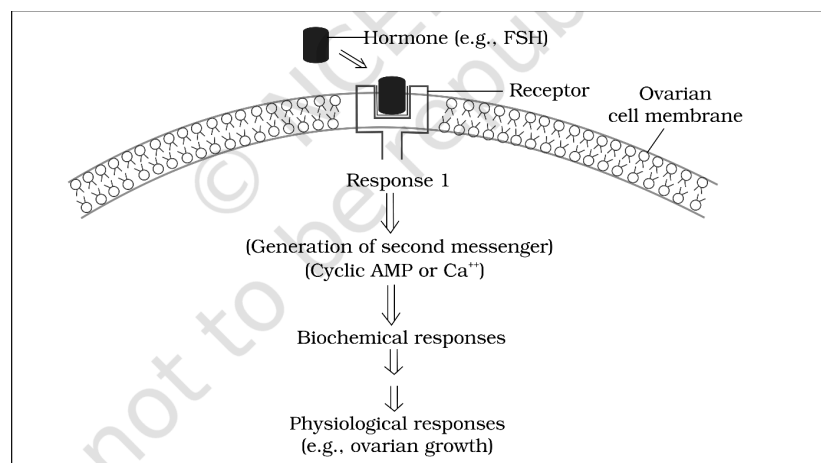


Figure 19.5 Diagrammatic representation of the mechanism of hormone action : (a) Protein hormone

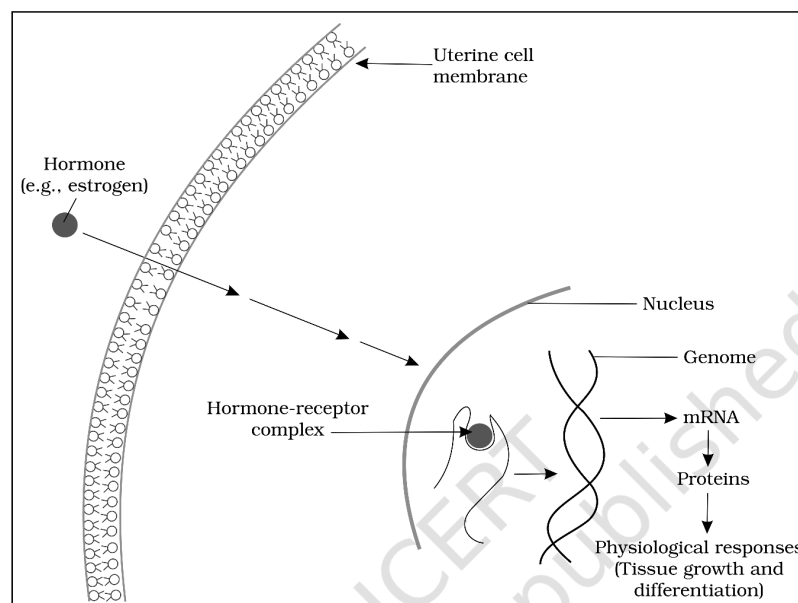


Figure 19.5 Diagrammatic representation of the mechanism of hormone action : (b) Steroid hormone

[NOTE] Reading Figure 19.5 (b): The steroid hormone (e.g., estrogen) enters through the **uterine cell membrane** into the **nucleus**, forming a **hormone-receptor complex** with intracellular receptors that interacts with the **genome** to produce mRNA and proteins, resulting in **physiological responses (tissue growth and differentiation)**.

[MEMORY AID - not in NCERT] Two mechanisms, one question: does the hormone **get in**? Protein/peptide hormones **stay outside** and shout through a **second messenger**; steroids and iodothyronines **walk in** and talk to the **genome**.
Water-soluble stays out, lipid-soluble goes in.

Recap

Quick recap of the whole chapter

- There are **special chemicals** which act as **hormones** and provide **chemical coordination, integration and regulation** in the human body; they **regulate metabolism, growth and development of our organs**.
- The **endocrine system** is composed of **hypothalamus, pituitary and pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads**; **gastrointestinal tract, kidney and heart** also produce hormones.
- **Pituitary**: **pars distalis** produces **six trophic hormones**; **pars intermedia** secretes **only one**; **pars nervosa** secretes **two**. The pituitary hormones regulate **somatic tissue growth** and **peripheral endocrine glands**.
- **Pineal**: **melatonin** governs the **24-hour (diurnal) rhythm** - sleep and waking, body temperature.
- **Thyroid**: **BMR, central neural system maturation, erythropoiesis, carbohydrate/protein/fat metabolism, menstrual cycle**; **thyrocalcitonin** lowers blood calcium. **Parathyroid**: **PTH** raises it.
- **Thymus**: **thymosins** differentiate **T-lymphocytes** (cell-mediated immunity) and raise **antibody** production (humoral immunity).
- **Adrenal**: **medulla** secretes **epinephrine and norepinephrine**; **cortex** secretes **glucocorticoids** and **mineralocorticoids**.
- **Pancreas**: **glucagon** raises blood glucose (glycogenolysis, gluconeogenesis); **insulin** lowers it (uptake, utilisation, glycogenesis). Failure of insulin gives **diabetes mellitus**.
- **Gonads**: **testis** secretes **androgens**; **ovary** secretes **estrogen and progesterone**.
- **Non-glands**: **ANF** from the heart lowers blood pressure, **erythropoietin** from the kidney drives RBC formation, and the **gastrointestinal tract** secretes **gastrin, secretin, cholecystokinin** and **gastric inhibitory peptide**.